

The graveyard of strategies

A survivorship audit of trading systems published in industry and academic literature, 1995–2010

Abstract

We construct a database of approximately 200 systematic trading strategies published in industry magazines (Stocks & Commodities, Futures Magazine, Active Trader) and academic journals between 1995 and 2010, then track each strategy's out-of-sample performance through 2018. The headline finding is that approximately 73% of published strategies have either been retired by their authors, ceased to produce positive returns post-publication, or generated returns statistically indistinguishable from zero. The post-publication decay is consistent with the data-snooping concerns of Sullivan, Timmermann, and White (1999), the post-publication anomaly decay documented by McLean and Pontiff (2016), and the limits-to-arbitrage framework of Shleifer and Vishny (1997). Strategy categories most prone to post-publication decay are short-window mean-reversion (90% failure rate) and high-frequency cross-sectional anomalies (82%); the most resilient categories are long-window trend-following (45% failure rate) and structurally-justified carry strategies (38%). We discuss the methodological challenges of strategy auditing, the implications for strategy publishing and disclosure, and what the failure pattern suggests about how strategies should be selected, tested, and communicated to end users.

Key findings

- i. **Roughly three-quarters of published strategies fail post-publication.** Of approximately 200 trading strategies published 1995–2010, 73% have either been retired by their authors or ceased producing positive out-of-sample returns by 2018.
- ii. **Failure rates vary sharply by strategy type.** Short-window mean-reversion strategies fail at 90%, high-frequency cross-sectional anomalies at 82%, but long-window trend-following at only 45% and structurally-justified carry at 38%.
- iii. **Decay is consistent with academic predictions.** Sullivan-Timmermann-White (1999) on data-snooping bias and McLean-Pontiff (2016) on post-publication decay both predict the magnitude of failure observed.

iv. **Disclosure practices in industry are systematically biased.** Most industry publications report strategies at peak performance, do not disclose the search space examined, and do not follow up on previously-published strategies that have failed.

v. **Survivorship bias in marketing materials is systematic.** Vendors selling strategies typically discontinue marketing of underperforming strategies without public retraction, leaving the impression of higher base success rates than the population supports.

EXEC Executive summary

Most published trading strategies do not work after they are published. The pattern is consistent enough across two decades and several hundred strategies that it constitutes a structural feature of the strategy-publishing ecosystem rather than coincidence.

This note constructs and analyses a database of approximately 200 systematic trading strategies published in industry magazines and academic journals between 1995 and 2010. For each strategy, we record the publication date, the asset class, the strategy category (mean-reversion, momentum, carry, trend-following, statistical arbitrage, etc.), the in-sample reported performance, and the strategy's status as of 2018. Strategies are classified as "alive" (still producing positive out-of-sample returns or actively traded by named practitioners), "decayed" (positive but materially reduced out-of-sample performance), "failed" (zero or negative out-of-sample performance), or "retired" (publicly discontinued by author or sponsoring firm).

The headline finding: approximately 27% of strategies are still alive in any meaningful sense by 2018; 73% have failed, decayed materially, or been retired. The pattern is consistent across asset classes and across publication venues, although the magnitudes differ by strategy category. Short-window mean-reversion strategies fail at the highest rate (90%); long-window trend-following and structurally-justified carry strategies fail at lower rates (45% and 38% respectively).

The finding is consistent with the academic literature on data-snooping bias and post-publication anomaly decay. [Sullivan, Timmermann, and White \(1999\)](#) showed that the apparent statistical significance of trading rules tested across multiple specifications collapses substantially when proper data-snooping corrections are applied. [McLean and Pontiff \(2016\)](#) documented that academic anomalies decline by approximately 26% in raw terms and 58% in t-statistic terms after publication. Our findings extend these to the broader population of industry-published strategies, which face less methodological scrutiny than academic anomalies and therefore exhibit more severe failure rates.

The implications are uncomfortable for the strategy-publishing ecosystem. Industry publications systematically report strategies at peak performance, do not disclose the search space examined to find the published strategy, and do not follow up on previously-published strategies that have failed. Vendors selling strategies typically discontinue marketing of failed strategies without public retraction, leaving the impression that the base rate of success is higher than it is. End users — practitioners and retail traders — encounter a heavily curated upper-tail sample without seeing the lower tail.

The note proceeds as follows. Section 2 reviews the relevant academic literature. Section 3 describes the database construction and audit methodology. Section 4 presents the empirical results. Section 5 examines the cross-sectional pattern by strategy category. Section 6 discusses implications for disclosure, publication, and strategy selection.

01 Introduction

The publication of new trading strategies is a continuous activity in financial-industry media and academic finance journals. Industry magazines have published trading systems and technical-indicator combinations as a primary editorial product for more than fifty years; academic journals publish hundreds of new asset-pricing anomalies each decade. The question of whether these strategies actually work — out of sample, after publication, in the hands of practitioners — has historically received less attention than the publication itself.

The academic literature has gradually come to address this question. [Lo and MacKinlay \(1990\)](#) introduced the concept of data-snooping bias in the context of asset-pricing tests, showing that apparent statistical significance can arise mechanically from testing many specifications on the same dataset. [Sullivan, Timmermann, and White \(1999\)](#) formalised this concern with a bootstrap-based test for technical trading rules and found that the apparent profitability of many rules disappears under proper correction. [McLean and Pontiff \(2016\)](#) tracked 97 academic anomalies post-publication and found systematic decay consistent with publication revealing previously-unexploited mispricing.

The industry-publication side of the question has been examined less systematically. [Bajgrowicz and Scaillet \(2012\)](#) tested 7,846 technical trading rules with appropriate multiple-comparisons corrections across 110 years of Dow Jones data, finding that essentially none survive the joint criteria of statistical significance and economic significance after transaction costs. [Aronson \(2007\)](#), working from a practitioner perspective, applied formal hypothesis testing to a large database of technical trading rules and arrived at similar conclusions.

This note contributes a more focused audit of the specific industry-publication universe — strategies that were presented as actionable to retail and professional practitioners through industry magazines and academic journals between 1995 and 2010. The fifteen-year window allows roughly 8–23 years of out-of-sample evidence per strategy, sufficient to distinguish genuine post-publication performance from noise. The 1995 starting point coincides with the maturation of widely-available historical datasets that made systematic backtesting routine; the 2010 ending point provides at least eight years of out-of-sample observation.

Three specific aims structure the analysis. First, to quantify the population-level failure rate of published strategies — the headline statistic. Second, to identify cross-sectional patterns: which strategy categories fail more often, which less. Third, to evaluate whether the observed failure pattern is consistent with the academic predictions from data-snooping and post-publication-decay theory.

02 Literature review

The relevant literature falls into three threads.

Data-snooping bias. When researchers test many strategy specifications on a single historical dataset, the best-performing specification will appear statistically significant by chance even if the underlying process is random. [Lo and MacKinlay \(1990\)](#) introduced the concern formally; [White \(2000\)](#) developed the "reality check" bootstrap test that controls family-wise error rates across multiple specifications. The implication is that strategies selected through extensive search exhibit in-sample performance that materially exceeds their true expected performance, and this gap is closed at publication when the strategy begins to be evaluated out-of-sample.

Post-publication decay. [McLean and Pontiff \(2016\)](#) tracked 97 academic asset-pricing anomalies through their post-publication histories and found that the average anomaly's return declines by approximately 26% post-publication, with the decline concentrated in the first few years. [Chordia, Subrahmanyam, and Tong \(2014\)](#) document similar decay for major U.S. equity anomalies. The mechanisms include arbitrage capital flowing into newly-disclosed anomalies (compressing the spreads), changes in market microstructure that alter the conditions under which the anomaly was originally documented, and pure data-snooping (the anomaly was never real to begin with, only spuriously identified in the original sample).

Limits to arbitrage and strategy persistence. [Shleifer and Vishny \(1997\)](#) developed the framework explaining why some anomalies persist post-publication despite arbitrage activity: institutional and contractual constraints prevent the marginal investor from exploiting them at scale. The framework predicts that anomalies with characteristics that limit arbitrage (high capacity constraints, high implementation costs, high risk-of-ruin during drawdowns) should decay less than those without such limits. We find empirical support for this in our cross-sectional results.

03 Database construction and methodology

3.1 Source population

We constructed our strategy database from four primary sources covering the 1995–2010 publication window:

- **Industry magazines:** Technical Analysis of Stocks & Commodities (TASC), Futures Magazine, Active Trader, Trader's Magazine. We catalogued all strategy articles that included specific entry and exit rules sufficient for replication.
- **Academic journals:** Journal of Finance, Journal of Financial Economics, Review of Financial Studies, Journal of Portfolio Management, Financial Analysts Journal. Strategy papers (as distinguished from pure-theory papers) were included.
- **Practitioner books:** Strategies that received book-length treatment from named authors (Murphy, Pring, Connors, Faith, Schwager) were included if specific rules were given.
- **Working paper repositories:** SSRN papers in the Trading and Markets category from the same window, where the paper proposed a specific tradeable strategy.

The full database contains 217 distinct strategies. We exclude 17 that we determined to be effectively duplicates of others (same rules, different authors), leaving 200 unique strategies for analysis.

3.2 Strategy classification

Table 1. Strategy database composition by category

Category	N	Share
Mean reversion (short window)	42	21%
Momentum / trend-following (long window)	38	19%
Cross-sectional equity anomalies	35	17.5%
Pattern recognition / chart formations	28	14%
Volatility / options strategies	19	9.5%
Carry / yield-based	14	7%
Statistical arbitrage / pairs	12	6%
Multi-asset / macro	12	6%

3.3 Status classification

For each strategy, we attempted to determine its status as of 2018 using the following hierarchy:

- **Alive:** The strategy can be replicated using the published rules and produces positive risk-adjusted returns over the post-publication period (publication date to 2018).
- **Decayed:** Strategy is replicable and still positive but with materially reduced performance (Sharpe ratio dropped by 50% or more relative to in-sample).
- **Failed:** Strategy is replicable but produces zero or negative risk-adjusted returns over the post-publication period.
- **Retired:** The strategy author or sponsoring firm has publicly discontinued, retracted, or substantially modified the strategy.

Replication and post-publication backtesting were performed using the rules as published, applied to standard data sources (Bloomberg, CRSP, and similar). Where strategies depended on proprietary data we were unable to access, we coded them as "unable to verify" and excluded them from the headline percentages — leaving 184 strategies in the verifiable subset.

04 Results

4.1 Headline failure rate

Table 2. Status of 184 verifiable strategies as of 2018

Status	N	Share
Alive	50	27.2%
Decayed	31	16.8%
Failed	68	37.0%
Retired	35	19.0%
Failure or decay (combined)	134	72.8%

Total strategies audited

184

Failure or decay rate

72.8%

"Alive" rate

27.2%

The headline statistic: approximately 27% of strategies in our sample are still alive in any meaningful sense, and 73% have failed, decayed, or been retired. The pattern is consistent with a structural feature of the strategy-publishing ecosystem rather than chance, given the sample size of 184 and the consistency across publication venues and asset classes.

4.2 Cross-sectional pattern

Table 3. Failure rates by strategy category

Category	N	Failure / decay rate
Mean reversion (short window)	42	90%
Cross-sectional equity anomalies	32	82%
Pattern recognition / chart formations	26	81%
Statistical arbitrage / pairs	11	73%
Volatility / options	17	71%
Multi-asset / macro	10	60%
Momentum / trend-following (long window)	36	45%
Carry / yield-based	10	38%

The cross-sectional pattern is informative. The categories with the highest failure rates are those most exposed to data-snooping concerns: short-window mean reversion (where the universe of possible parameter combinations is enormous and many spurious specifications can be found), cross-sectional equity anomalies (where post-publication decay is most-documented), and pattern recognition (where the visual appeal of a "pattern" can mask the absence of statistical significance after multiple-comparisons correction).

The categories with the lowest failure rates are those with structural justification grounded outside the historical sample: long-window trend-following (which has theoretical support from the time-series momentum literature and persistent

behavioural underreaction stories) and carry strategies (where the premium has cross-asset universality and a credible risk-premium interpretation).

The pattern is consistent with the limits-to-arbitrage prediction that strategies with greater structural grounding should decay less. Strategies whose only justification is their in-sample performance face the full force of data-snooping; strategies whose justification rests partly on theory or behavioural mechanism have something to fall back on when in-sample fits begin to deteriorate.

4.3 Time-since-publication pattern

Failure rates also vary by time since publication. Strategies published 1995–2000 have a combined failure-or-decay rate of 84%; those published 2005–2010 have a rate of 65%. The difference reflects two effects: (a) longer post-publication windows allow more failures to materialise, and (b) more recent publications may include strategies that will fail in the future but have not yet done so.

Table 4. Failure rate by publication era

Publication era	N	Failure / decay rate
1995–2000	62	84%
2001–2005	68	75%
2006–2010	54	65%

Extrapolating, we should expect that strategies published in 2015–2018 (which we cannot yet fully audit) will eventually exhibit failure-or-decay rates approaching 80%, consistent with the historical pattern. The failure pattern does not appear to be improving over time as the literature on data-snooping has become more widely known, suggesting that publication incentives outweigh methodological rigour in driving the population-level outcome.

05 Implications and discussion

5.1 The disclosure problem

The structural failure rate of published strategies is well-known to academics but poorly known to retail practitioners. The asymmetry has direct consequences. A practitioner encountering a strategy in an industry magazine sees one strategy, presented at peak performance, with no indication of what failed before it. The relevant base rate — the probability that any given published strategy will work post-publication — is approximately 27%; the practitioner has no way to know this from the article alone.

The asymmetry is exacerbated by vendor practices. A vendor selling a strategy typically markets it during periods of strong performance and discontinues marketing during periods of poor performance. Strategies that fail are quietly retired without retraction; new strategies replace them in the marketing materials. The base-rate evidence available to a retail practitioner therefore systematically over-represents successful strategies and under-represents failed ones.

The honest disclosure response is straightforward: when a strategy is published or sold, the failure history of similar strategies should be disclosed. The vendor should publicly retain previously-failed strategies in their public record rather than removing them from view. The base rate of success in the broader strategy population should be communicated, not the success rate within the seller's curated subset.

5.2 What characterises the survivors

The 27% of strategies still alive in our sample share several characteristics:

- **Structural justification.** They have theoretical or behavioural grounding beyond in-sample fit — most commonly behavioural-finance mechanisms (underreaction, anchoring, disposition effects) or risk-premium mechanisms (carry, term premium, illiquidity).
- **Long window parameters.** They use parameters at horizons of months or longer rather than days. Short-window strategies have many more degrees of freedom for spurious in-sample fitting.
- **Limited capacity.** They are subject to capacity constraints that limit arbitrage activity (small-cap value, microstructure-dependent strategies). Large-capacity strategies face faster decay.
- **Cross-asset replication.** They have been documented in multiple asset classes or markets, reducing the probability of pure single-market data-snooping.
- **Imperfect implementation.** They are difficult to implement perfectly due to operational complexity, transaction costs, or governance constraints — providing a "moat" that prevents arbitrage from fully eliminating the premium.

The composite picture of a "likely-to-survive" strategy is not glamorous: a long-window strategy, grounded in behavioural or risk-premium theory, with limited capacity and operational difficulty. The strategies that look most attractive in marketing materials — high Sharpe ratios, frequent trades, simple rules, broad applicability — are systematically those most likely to fail.

5.3 Conclusion

Approximately 73% of trading strategies published in the 1995–2010 window have failed, decayed, or been retired by 2018. The failure pattern is consistent with academic predictions from data-snooping bias and post-publication decay theory. The cross-sectional pattern — high failure rates for short-window mean-reversion and pattern-recognition strategies, lower failure rates for long-window trend-following and carry — is consistent with the limits-to-arbitrage framework.

The implications are direct. For practitioners, the relevant base rate of strategy success is approximately 27%, not the near-100% that marketing materials imply. For publishers and vendors, transparent disclosure of failed strategies is the only honest practice; quiet retirement of failed strategies misrepresents the population. For strategy selection, the

characteristics of survivors — structural justification, long windows, limited capacity, cross-asset replication — provide a checklist that materially improves ex-ante selection probability.

Limitations. Our database is necessarily incomplete; the strategies we have catalogued are themselves a curated subset of the population, since strategies published in obscure venues or never widely distributed are under-represented. Our status classification involves judgement calls (the boundary between "decayed" and "failed" is subjective). Our post-publication backtesting uses standard data sources that may not perfectly replicate the data the original author used. The 2018 cutoff is somewhat arbitrary; longer post-publication windows would likely produce higher failure rates as more strategies fail with time.

The base rate of success for a published trading strategy is 27%, not the near-100% implied by marketing materials. The honest version of strategy publishing requires keeping the failures visible.

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APX Appendix A: Methodology details and selected case studies

A.1 Replication procedure

For each strategy in our database, we attempted to reconstruct the published rules in code, applying them to standard data sources (Bloomberg historical data for institutional sources, CRSP for academic sources, and Yahoo/Quandl for retail-magazine sources where higher-cost data was not used by the original author). Where the original author specified a particular data vendor whose data we could not access, we used the closest available substitute and noted the data substitution.

Backtests were run from publication date to December 2017, providing a minimum of 7 years and a maximum of 22 years of out-of-sample evidence. Performance metrics were computed on a like-for-like basis with the original publication where possible; where the original publication used non-standard metrics, we re-ran the standard set (CAGR, max drawdown, Sharpe ratio, time-in-market) for consistency.

A.2 Selected case studies

Case 1: A short-window mean-reversion strategy. Published in TASC 1998, the strategy used a 3-day RSI with specific overbought/oversold thresholds applied to the S&P 500. In-sample (1990–1997) performance was reported as 18% annualised at low drawdown. Post-publication (1998–2017): 4% annualised, materially below buy-and-hold for the period and below T-bills for substantial sub-periods. Status: failed.

Case 2: A long-window trend-following strategy. Published in Journal of Portfolio Management 2003, the strategy used a 200-day moving average filter on a multi-asset basket. In-sample (1973–2002) Sharpe ratio reported as 0.7. Post-publication (2003–2017): Sharpe ratio of 0.6, modestly degraded but qualitatively similar. Status: alive.

Case 3: A high-frequency cross-sectional anomaly. Published in Journal of Finance 2005, identifying a short-window predictability pattern in cross-sectional equity returns. In-sample t-statistic exceeded 5. Post-publication: t-statistic dropped to 1.2 over the 2005–2017 window. Status: failed (consistent with the McLean-Pontiff post-publication-decay finding).

Case 4: A volatility arbitrage strategy. Published in a 2007 trader's book, exploiting a specific structural feature of VIX futures. In-sample backtest reported 25% annualised. Post-publication: positive but volatile, with substantial drawdowns in 2008 and 2018. Status: decayed (still alive but materially weaker than published).

A.3 Cross-sectional regression

We regressed an indicator for "alive at 2018" on strategy characteristics: log of in-sample Sharpe ratio, indicator for short-window (≤ 30 days), indicator for academic-vs-industry publication

A.4 ProRealTime: walk-forward audit template

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Walk-forward backtest template for strategy auditing
// Use to systematically test post-publication out-of-sample performance

DEFPARAM CumulateOrders = False

// In-sample / out-of-sample split – typically by date
publicationBar = 2500 // bar index of strategy publication

// In-sample: bars 1 to publicationBar (the strategy author's data window)
```

```

// Out-of-sample: bars publicationBar+1 to current

// The audit principle: report metrics SEPARATELY for IS and OOS
// Do not report combined-period metrics – they obscure post-publication decay

// Example signal: parameterised by the published rule
// (replace with the actual published rule for the strategy being audited)
fastMA = Average[10](close)
slowMA = Average[50](close)

IF fastMA > slowMA AND NOT LongOnMarket THEN
    BUY 1 CONTRACT AT MARKET
ENDIF

IF fastMA < slowMA AND LongOnMarket THEN
    SELL AT MARKET
ENDIF

// In post-processing: compare IS Sharpe to OOS Sharpe.
// If OOS Sharpe < 50% of IS Sharpe, classify as "decayed."
, indicator for asset class, and indicator for behavioural/risk-premium structural justification. The strongest predictors
of survival are: long-window indicator (positive coefficient,  $t = 4.2$ ) and structural justification indicator (positive,  $t = 3.8$ ). In-sample Sharpe is weakly negative (higher in-sample Sharpe predicts higher failure probability, consistent
with the data-snooping interpretation), with  $t = -1.9$ .

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